AQUÍ VA EL TÍTULO DEL ARTÍCULO

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ABSTRACT. Aquí va el resumen

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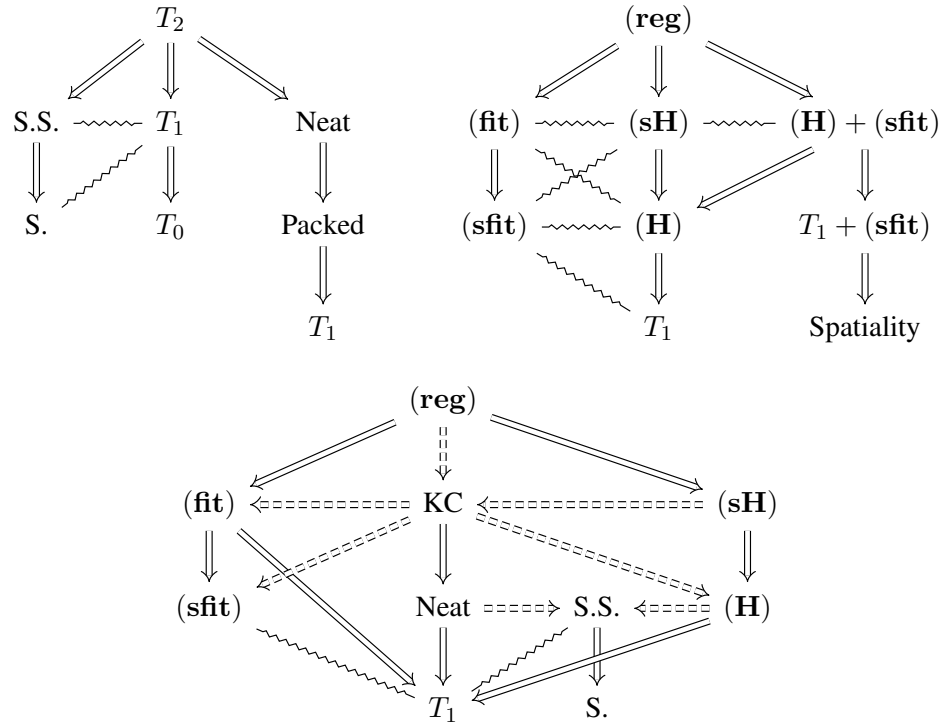
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1. INTRODUCTION

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2. PRELIMINARIES

2.1. **Frames, nuclei, and open filters.** We recall the basic notions and notation concerning frames, nuclei, and open filters; for general background, see the standard references on point-free topology.

A *frame* is a complete lattice A satisfying

$$a \wedge \bigvee X = \bigvee \{a \wedge x \mid x \in X\}.$$

A frame homomorphism preserves finite meets and arbitrary joins. We write $\mathbf{Frm}$ for the category of frames. The Heyting implication on A is characterized by

$$x \wedge a \leq b \quad \Longleftrightarrow \quad x \leq (a \succ b).$$

Every frame homomorphism $f : A \to B$ has a right adjoint $f_* : B \to A$, characterized by $f(a) \leq b$ if and only if $a \leq f_*(b)$.

A *nucleus* on A is a map $j : A \to A$ satisfying

$$x \leq j(x), \qquad j(j(x)) = j(x), \qquad j(x \wedge y) = j(x) \wedge j(y).$$

The fixed-point set $A_j = \{x \in A \mid j(x) = x\}$ is a frame, with joins given by $\bigvee_{A_j} X = j(\bigvee_A X)$. The map $j^* : A \to A_j$, $j^*(x) = j(x)$, is a surjective frame homomorphism, and every quotient arises in this way. We denote by NA the frame of nuclei on A. For $a \in A$, put

$$u_a(x) = a \vee x, \qquad v_a(x) = a \succ x, \qquad w_a(x) = (x \succ a) \succ a.$$

The nuclei u_a and v_a determine the closed and open quotients, respectively.

An element $p \in A$ is $\wedge$-*irreducible* if $p \neq 1$ and $x \wedge y \leq p$ implies $x \leq p$ or $y \leq p$. The set $\mathrm{pt}(A)$ of such elements is endowed with the topology generated by

$$\mathcal{U}(a) = \{p \in \mathrm{pt}(A) \mid a \not\leq p\}, \qquad a \in A.$$

The assignment $\mathcal{U} : A \to \mathcal{O}(\mathrm{pt}(A))$ is a surjective frame homomorphism, called the *spatial reflection*. A frame is *spatial* when $\mathcal{U}$ is an isomorphism. Points correspond bijectively to completely prime filters through $F_p = \{a \in A \mid a \not\leq p\}$.

A filter F on A is *open* if, for every directed $D \subseteq A$,

$$\bigvee D \in F \quad \Longrightarrow \quad D \cap F \neq \varnothing.$$

We write $A^\wedge$ for the collection of open filters. If $f : A \to B$ is a frame homomorphism with right adjoint f_*, then

$$G \in B^\wedge \Longrightarrow f_*G \in A^\wedge, \qquad f_*G = \{a \in A \mid f(a) \in G\}. \tag{1}$$

Proposition 2.1. *Let $f = f^* \colon A \to B$ be a frame morphism. If $G \in B^\wedge$, then $f_*G \in A^\wedge$.*

Every nucleus j determines its *admissible filter*

$$\nabla(j) = \{a \in A \mid j(a) = 1\}.$$

Given a filter F, the least nucleus admitting F is

$$v_F = f_F^\infty = \bigvee_{a \in F} v_a, \qquad f_F(x) = \bigvee_{a \in F} (a \succ x), \tag{2}$$

where the join defining v_F is taken in NA. For $F \in A^\wedge$, one has $\nabla(v_F) = F$. Thus, open filters correspond bijectively to fitted nuclei whose admissible filters are open.

The Hofmann–Mislove theorem identifies $A^\wedge$ with the compact saturated subsets of $\mathrm{pt}(A)$. Explicitly, the corresponding sets and filters are

$$Q_F = \{p \in \mathrm{pt}(A) \mid (\forall a \in F)\ a \not\leq p\}, \qquad F_Q = \{a \in A \mid Q \subseteq \mathcal{U}(a)\}.$$

We refer to (F, Q_F) as an ***HM pair***. When $\mathrm{pt}(A) = \varnothing$, the only open filter is A, corresponding to $Q_A = \varnothing$ and to the trivial fitted quotient $v_A = \top_{NA}$.

2.2. **Compact quotients and admissibility intervals.** Recall that the quotient A_j determined by a nucleus j is compact if and only if $\nabla(j)$ is an open filter. In particular, for each $F \in A^\wedge$, the fitted nucleus v_F determines a compact quotient. However, a compact quotient need not be uniquely determined by its admissible filter.

For $F \in A^\wedge$, let

$$M_F = \{m \in A \setminus F \mid m \text{ is maximal in } A \setminus F\}.$$

Define

$$w_F(x) = \bigwedge \{m \in M_F \mid x \leq m\}$$

and the nucleus w_F is the greatest nucleus with admissible filter F, whereas v_F is the least one. Consequently,

$$\{j \in NA \mid \nabla(j) = F\} = [v_F, w_F].$$

We call this the *admissibility interval* associated with F. Evaluating at 0 yields

$$[v_F(0), w_F(0)] \subseteq A.$$

If $\mathrm{pt}(A)$ is T_1, then $M_F = Q_F$, identifying points with their corresponding $\wedge$-irreducible elements.

We shall also use nuclei induced by subspaces. For a topological space S and $E \subseteq S$, define

$$[E] : \mathcal{O}S \longrightarrow \mathcal{O}S, \qquad [E](U) = (E \cup U)^\circ.$$

A nucleus of this form is called *spatially induced*.

Let Q be a compact saturated subset of S, put $G = \{U \in \mathcal{O}S \mid Q \subseteq U\}$, and write $Q' = S \setminus Q$. Then

$$\nabla([Q']) = G, \qquad v_G \leq [Q'] \leq w_G.$$

More generally, let $F \in A^\wedge$, let $Q = Q_F$, and put $G = F_Q = \mathcal{V}(Q)$, viewed as an open filter of $\mathcal{O}(\mathrm{pt}(A))$. We call (F, Q, G) an *HM triple*. Denote by

$$\mathcal{U}_* : \mathcal{O}(\mathrm{pt}(A)) \to A$$

the right adjoint of the spatial reflection.

Proposition 2.2. *For every $k \in [v_G, w_G]$, the composite*

$$\mathcal{U}_* \circ k \circ \mathcal{U}$$

is a nucleus on A with admissible filter F.

Hence Proposition 2.5 gives the map

$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

and, in particular,

$$v_F \leq \Phi(k) \leq w_F.$$

Finally, when $A = \mathcal{O}S$, identifying S with $\mathrm{pt}(A)$, the spatial reflection is an isomorphism.

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Lemma 2.3. *If $A = \mathcal{O}S$, then $\mathcal{U}_* \circ [E] \circ \mathcal{U} = [E]$ for every $E \subseteq S$.*

3. Separation properties induced by compact quotients

The notions studied in this section are motivated by the work of Sexton and Simmons on point-sensitive and point-free patch constructions. Our aim is to identify nuclei associated with open filters that determine compact closed quotients, and to investigate the separation properties arising from this requirement. We begin by recalling Sexton's notion of neatness ([6, Definition 4.3] and [8, Definition 8.2.1]), which provides the starting point for our approach.

Definition 3.1. Let A be a frame, let $F \in A^\wedge$ and $\alpha \in \mathrm{Ord}$. We say that:

- F is α-neat, if

$$x \vee d = 1, \quad \text{for every } x \in F,$$

  where $d = f_F^{\alpha}(0)$.
- A is α-neat if every $F \in A^{\wedge}$ is α-neat.
- A is neat if it is α-neat for some α.

Let us record a useful consequence of neatness. Suppose that A is α-neat and let $F \in A^{\wedge}$. Put $d = f_F^{\alpha}(0)$. Since $x \vee d = 1$ for every $x \in F$, we have $F \subseteq \nabla(u_d)$. Thus, $f_F \leq u_d$, and consequently

$$v_F = f_F^{\infty} \leq u_d.$$

On the other hand,

$$d = f_F^{\alpha}(0) \leq v_F(0),$$

so that

$$u_d \leq u_{v_F(0)} \leq v_F.$$

Therefore,

$$v_F = f_F^{\infty} = u_d.$$

In particular, every fitted nucleus associated with an open filter is closed, and hence every such compact quotient is simultaneously closed.

Sexton showed that a topological space is neat if and only if it is both packed and stacked [6, Theorem 6.2]. Motivated by this spatial characterization, we now introduce point-free counterparts of stackedness and strong stackedness.

Let S be a topological space, let $Q \subseteq S$ be compact saturated, and let

$$G = \mathcal{V}(Q) \in (\mathcal{O}S)^{\wedge}.$$

Recall that S is stacked if

$$v_G(\varnothing) = (Q')^{\circ}$$

for every such pair (G, Q), and strongly stacked if

$$v_G = [Q']$$

for every such pair.

We now return to the general setting. Let A be a frame, $S = \mathrm{pt}(A)$, and let (F, Q, G) be an H–M triple. Recall that the map of Proposition 2.2

$$\Phi : [v_G, w_G] \longrightarrow [v_F, w_F], \qquad \Phi(k) = \mathcal{U}_* \circ k \circ \mathcal{U},$$

provides a natural way of comparing the admissibility interval associated with G with that associated with F.

Proposition 3.2. *Let (F, Q, G) be an H–M triple. Then $\Phi(w_G) = w_F$, that is,*

$$w_F = \mathcal{U}_* \circ w_G \circ \mathcal{U}.$$

Proof. By Proposition 2.2, we have

$$\Phi(w_G) = U_* \circ w_G \circ U \leq w_F.$$

For the reverse inequality, by adjunction it suffices to show that

$$U \circ w_F \leq w_G \circ U.$$

Since $w_G = [M']$, for every $x \in A$ we have

$$w_G(U(x)) = [M'](U(x)) = (M' \cup U(x))^\circ.$$

Thus, it is enough to prove that

$$U(w_F(x)) \subseteq M' \cup U(x).$$

Suppose that $y \in U(w_F(x))$ and $y \notin M'$. Since $y \notin M'$, we have $y \in M$. Moreover, $y \in U(w_F(x))$ implies $w_F(x) \not\leq y$. But

$$w_F(x) = \bigwedge\{m \in M \mid x \leq m\}.$$

Hence $x \not\leq y$, since otherwise $w_F(x) \leq y$, a contradiction. Therefore $y \in U(x)$, proving the desired inclusion. Consequently, $\Phi(w_G) = w_F$. □

Proposition 3.2 shows that Φ preserves the upper elements of the corresponding admissibility intervals. Since $[Q']$ is the spatially induced nucleus appearing in Sexton's definitions, the nucleus

$$\Phi([Q']) = U_* \circ [Q'] \circ U$$

provides a natural point-free counterpart. This motivates the following definitions.

Definition 3.3. Let A be a frame and (F, Q) an H–M pair. We say that:

- A is *FQ-stacked* if
$$(U_* \circ [Q'] \circ U)(0) = v_F(0).$$
- A is *FQ-strongly stacked* if
$$U_* \circ [Q'] \circ U = v_F.$$

The frame A is *stacked* (respectively, *strongly stacked*) if it is FQ-stacked (respectively, FQ-strongly stacked) for every H–M pair (F, Q).

Clearly, every strongly stacked frame is stacked. Let us look that these properties are conservative. We first establish the relation between the transfinite iterations defining the fitted nuclei on a space and on its frame of open sets.

Lemma 3.4. *Let $\alpha \in \mathrm{Ord}$ and $X \in CS$. Then*

$$\partial_F^\alpha(X) = \big(f_F^\alpha(X')\big)'.$$

In particular, for every $U \in \mathcal{O}S$,

$$\big(\partial_F^\infty(U')\big)' = v_F(U).$$

Proof. We proceed by transfinite induction on α. For $\alpha = 0$, the identity follows from $\partial_F^0 = \mathrm{id} = f_F^0$.

Suppose that the identity holds for α. Then

$$\begin{aligned}\partial_F^{\alpha+1}(X) &= \partial_F(\partial_F^\alpha(X)) \\ &= \big(f_F((\partial_F^\alpha(X))')\big)' \\ &= \big(f_F(f_F^\alpha(X'))\big)' \\ &= \big(f_F^{\alpha+1}(X')\big)'.\end{aligned}$$

If λ is a limit ordinal, then

$$\begin{aligned}\partial_F^\lambda(X) &= \bigcap_{\alpha<\lambda} \partial_F^\alpha(X) \\ &= \bigcap_{\alpha<\lambda} (f_F^\alpha(X'))' \\ &= \left(\bigcup_{\alpha<\lambda} f_F^\alpha(X')\right)' \\ &= \big(f_F^\lambda(X')\big)'.\end{aligned}$$

Thus, the identity holds for every ordinal α. Taking the stable value of the iterations yields the last assertion. □

Proposition 3.5. *The notions of stackedness and strong stackedness are conservative.*

Proof. Let S be a topological space and put $A = \mathcal{O}S$. For every H–M pair (F, Q), we have $v_F = [Q']$ whenever S is strongly stacked. By Lemma 2.3,

$$U_* \circ [Q'] \circ U = [Q'] = v_F.$$

Thus, A is strongly stacked.

Conversely, if A is strongly stacked, then

$$v_F = U_* \circ [Q'] \circ U = [Q']$$

for every H–M pair (F, Q), and hence S is strongly stacked.

Now suppose that S is stacked. By definition,

$$Q = \partial_F^\infty(S).$$

Consequently,

$$Q' = (\partial_F^\infty(S))' = v_F(\varnothing),$$

where the last equality follows from Lemma 3.4. By Lemma 2.3,

$$\begin{aligned}(U_* \circ [Q'] \circ U)(\varnothing) &= [Q'](\varnothing) \\ &= (Q')^\circ \\ &= Q' \\ &= v_F(\varnothing).\end{aligned}$$

Therefore, A is stacked.

Conversely, if A is stacked, then

$$v_F(\varnothing) = (U_* \circ [Q'] \circ U)(\varnothing) = Q'.$$

By Lemma 3.4,

$$(\partial_F^\infty(S))' = Q',$$

and hence $\partial_F^\infty(S) = Q$. Thus, S is stacked. □

Neatness requires every fitted nucleus associated with an open filter to determine a closed compact quotient. A natural extension is to impose this condition on every nucleus admitting an open filter. This leads to a point-free counterpart of Wilansky's classical KC property [11].

Definition 3.6. Let A be a frame. We say that A is *KC* if every compact quotient of A is closed.

In the spatial setting, Wilansky's KC property coincides with Sexton's notion of packedness. Consequently,

$$\text{Neat} \iff \text{KC and Stacked}.$$

Whether analogous relationships hold in the point-free setting will be addressed in Section **??**. We first examine how these properties interact with classical separation axioms.

3.1. **Comparison with Classical Separation Axioms.** Although stackedness and strong stackedness are conservative, neither property is comparable with the T_1 separation axiom, as the following examples show.

Example 3.7.

(1) Consider the frame $A = [0, 1]$ with its usual order. Its open filters are

$$F_r = (r, 1], \qquad 0 \le r < 1,$$

and the corresponding compact saturated subsets of $\operatorname{pt} A = [0, 1)$ are $Q_r = [0, r]$. A direct computation gives

$$v_{F_r}(x) = \begin{cases} x, & x \le r, \\ 1, & x > r, \end{cases} = (U_* \circ [Q_r'] \circ U)(x).$$

Thus, A is strongly stacked but not T_1.

(2) The Hausdorff frame of Paseka and Šmarda [4], considered in Example 3.16, is T_1 but not stacked, and hence not strongly stacked.

Recall that every T_1 stacked space is sober and that, in the strongly stacked case, $w_G = v_G$. We now investigate analogous phenomena for frames.

Proposition 3.8. *Let A be a frame. If*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then every point of A is maximal in $\operatorname{pt} A$.

Proof. Let $p, q \in \operatorname{pt} A$ with $p \leq q$, and consider the open filter

$$F_q = \{x \in A \mid x \not\leq q\}.$$

By the definition of w_{F_q},

$$w_{F_q} = v_q, \qquad \text{so} \qquad w_{F_q}(0) = q.$$

By hypothesis,

$$q = v_{F_q}(0) = \bigvee_{x \in F_q} \neg x.$$

For every $x \in F_q$, we have $x \not\leq q$, and hence $x \not\leq p$. Since p is a point,

$$x \wedge \neg x = 0 \leq p \quad \Longrightarrow \quad \neg x \leq p.$$

Consequently,

$$q = \bigvee_{x \in F_q} \neg x \leq p.$$

Thus $p = q$, proving that every point of A is maximal. □

Proposition 3.9. *Let A be a T_1 frame. Then:*

(1) *A is stacked if and only if*

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$.

(2) *A is strongly stacked if and only if*

$$v_F = w_F$$

for every $F \in A^\wedge$.

Proof. Let (F, Q) be an H–M pair. Since A is T_1, we have $Q = M$, and hence $w_G = [Q']$. By Proposition 3.2,

$$\mathcal{U}_* \circ [Q'] \circ \mathcal{U} = w_F.$$

Both assertions now follow directly from the definitions of stackedness and strong stackedness. □

Remark 3.10. Notice that the T_1 assumption is not needed to obtain stackedness from the equality

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$. Indeed, for every H–M pair (F, Q), we have

$$v_F \leq \mathcal{U}_* \circ [Q'] \circ \mathcal{U} \leq w_F.$$

Evaluating at 0 yields

$$v_F(0) \leq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) \leq w_F(0).$$

Thus, the equality of the endpoints implies that A is stacked.

The preceding characterizations assume that the frame is T_1. We now consider conditions under which the corresponding identities imply the T_1 separation property.

Corollary 3.11. *Let A be a frame such that*

$$\mathcal{V}(a) = \{p \in \operatorname{pt} A \mid a \leq p\} \neq \varnothing$$

for every $a \neq 1$. If

$$v_F(0) = w_F(0)$$

for every $F \in A^\wedge$, then A is T_1. In particular, the same conclusion holds if $v_F = w_F$ for every $F \in A^\wedge$.

Proof. By Proposition 3.8, every point of A is maximal in $\operatorname{pt} A$. Suppose that there exist $p \in \operatorname{pt} A$ and $a \in A$ such that $p < a < 1$. By hypothesis, there is $q \in \operatorname{pt} A$ with $a \leq q$. Hence

$$p < a \leq q,$$

contradicting the maximality of p in $\operatorname{pt} A$. Therefore, every point of A is maximal in A, and hence A is T_1. The last assertion follows immediately from $v_F = w_F$. □

While the strongly stacked condition yields the collapse of every admissibility interval, the stacked condition already guarantees this phenomenon for those associated with completely prime filters.

Corollary 3.12. *Let A be a T_1 stacked frame. Then every admissibility interval associated with a completely prime filter collapses. More precisely, for every $p \in \operatorname{pt} A$,*

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

Proof. Let $p \in \operatorname{pt} A$. Since A is T_1 and stacked, by Proposition 3.9 we have

$$v_{F_p}(0) = w_{F_p}(0) = p.$$

Moreover, since p is maximal in A, we have

$$w_{F_p} = u_p.$$

For every nucleus j on A, one has $u_{j(0)} \leq j$. Applying this to v_{F_p}, we obtain

$$u_p = u_{v_{F_p}(0)} \leq v_{F_p} \leq w_{F_p} = u_p.$$

Consequently,

$$[v_{F_p}, w_{F_p}] = \{u_p\}.$$

□

Notice that the additional hypothesis in Corollary 3.11 is automatically satisfied whenever A is spatial. Indeed, if $a < 1$, spatiality ensures the existence of $p \in \operatorname{pt} A$ such that $a \leq p$. Thus, in the spatial setting, our results recover the classical fact that every T_1 stacked space is sober.

The point-free situation is substantially different: even the stronger combination of T_1 and strong stackedness does not imply spatiality, as the following example shows.

Example 3.13. Let B be an atomless complete Boolean algebra and consider the frame

$$A = B \times 2,$$

where $2 = \{0 < 1\}$. Since B has no points and points of a finite product of frames correspond to points of its factors, we have

$$\operatorname{pt} A = \{p\}, \qquad p = (1, 0).$$

Clearly, p is maximal in A, so A is T_1. However, A is not spatial, since for distinct $a, b \in B$,

$$(a, 0) \neq (b, 0), \qquad \mathcal{U}(a, 0) = \mathcal{U}(b, 0) = \varnothing.$$

We now show that A is strongly stacked. Since B is an atomless complete Boolean algebra, its only compact element is 0. Thus,

$$K(A) = K(B) \times K(2) = \{(0, 0), (0, 1)\}.$$

As every open filter of a complete Boolean algebra is principal and generated by a compact element, it follows that

$$A^\wedge = \{A, F_p\}, \qquad F_p = B \times \{1\}.$$

Since $F_p = {\uparrow}(0, 1)$ and A is a complete Boolean algebra, we have

$$v_{F_p} = v_{(0,1)} = u_p.$$

Moreover, p is the unique maximal element of $A \setminus F_p$, so

$$w_{F_p} = u_p.$$

Consequently,

$$v_{F_p} = w_{F_p} = u_p.$$

For $F = A$,

$$v_A = \operatorname{tp} = w_A.$$

Then,

$$v_F = w_F \qquad \text{for every } F \in A^\wedge.$$

By Proposition 3.9, A is strongly stacked.

The preceding example yields the following result,

Theorem 3.14. *There exist non-spatial T_1 strongly stacked frames. In particular, there exist non-spatial T_1 stacked frames.*

Recall that $A^\wedge$ carries the topology generated by the basic open sets

$$D(x) = \{F \in A^\wedge \mid x \in F\}, \qquad x \in A.$$

Moreover, the points of the Vietoris modification VA (view [3]), are represented by pairs

$$(F, a), \qquad F \in A^\wedge, \quad a \in [v_F(0), w_F(0)].$$

Consider the continuous surjection

$$\pi : \operatorname{pt}(VA) \longrightarrow A^\wedge, \qquad (F, a) \longmapsto F.$$

It induces the frame morphism

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA)).$$

Recall also that the subbasic open sets of $\mathrm{pt}(VA)$ are determined by

$$(F, a) \in \Box x \iff x \in F, \qquad (F, a) \in \Diamond x \iff x \not\leq a. \tag{3}$$

Particularly,

$$a \in [F](x) \iff x \in F \quad \text{and} \quad a \in \langle F\rangle(x) \iff x \not\leq a. \tag{4}$$

Theorem 3.15. *Let A be a frame, and consider the continuous surjection*

$$\pi \colon \mathrm{pt}(VA) \longrightarrow A^\wedge, \qquad (F, a) \longmapsto F.$$

Then the following statements hold:

(1) *If A is T_1 and stacked, then*

$$\mathcal{O}(\pi) = \pi^{-1} \colon \mathcal{O}(A^\wedge) \longrightarrow \mathcal{O}(\mathrm{pt}(VA))$$

is an isomorphism.

(2) *If $\mathcal{O}(\pi)$ is an isomorphism, then A is stacked.*

(3) *If $\mathcal{O}(\pi)$ is an isomorphism and*

$$\mathcal{V}(a) = \{p \in \mathrm{pt}\, A \mid a \leq p\} \neq \varnothing$$

for every $a < 1$, then A is T_1.

Proof.

(1) Suppose that A is T_1 and stacked. By Proposition 3.9,

$$v_F(0) = w_F(0) \qquad (F \in A^\wedge).$$

Thus, every fiber of π is a singleton, so π is bijective.

It remains to show that π^{-1} is continuous. Recall that the subbasic open sets of $\mathrm{pt}(VA)$ are

$$[-](x) = \{(F, a) \mid x \in F\}$$

and

$$\langle - \rangle(x) = \{(F, a) \mid x \not\leq a\}.$$

For the first family,

$$\pi([-](x)) = D(x).$$

For the second, since $a = v_F(0)$ and $v_F(0) = w_F(0)$, we have

$$\begin{aligned} (F, a) \in \langle - \rangle(x) &\iff x \not\leq v_F(0) \\ &\iff x \not\leq w_F(0) \\ &\iff x \in F. \end{aligned}$$

Hence,

$$\pi(\langle - \rangle(x)) = D(x).$$

Therefore, π^{-1} is continuous, and π is a homeomorphism. Consequently, $\mathcal{O}(\pi)$ is an isomorphism.

(2) Suppose that $\mathcal{O}(\pi)$ is an isomorphism. Then π is injective. Since both
$$(F, v_F(0)), \qquad (F, w_F(0))$$
belong to $\mathrm{pt}(VA)$ and have the same image under π, we obtain
$$v_F(0) = w_F(0)$$
for every $F \in A^\wedge$. By Remark 3.10, A is stacked.

(3) Suppose additionally that
$$\mathcal{V}(a) \neq \varnothing \qquad (a < 1).$$
By part (2),
$$v_F(0) = w_F(0) \qquad (F \in A^\wedge).$$
By Corollary 3.11, it follows that A is T_1.

□

We now compare stackedness with the stronger separation property (H). As shown in [4], (H) implies T_1. However, (H) does not imply stackedness.

Example 3.16. Let $I = [0, 1]$ with its usual topology and put $A = \mathcal{O}(I)$. Following the construction of Paseka and Šmarda [4, Proposition 2.3], consider
$$\widetilde{A} = \{(u, v) \in A \times A_{\neg\neg} \mid u \leq v\},$$
where $A_{\neg\neg}$ is the Boolean algebra of regular elements of A.

By [4, Propositions 2.3 and 2.4], $\widetilde{A}$ is a compact frame satisfying (H), with
$$\mathrm{pt}\,\widetilde{A} = \{(I \setminus \{x\}, I) \mid x \in I\}.$$

Since $\widetilde{A}$ is compact, $F = {\uparrow}\, 1 = \{1\}$ is an open filter. Let (F, Q) be its corresponding H–M pair. As $1 \notin \mathrm{pt}\,\widetilde{A}$, we have $Q = \mathrm{pt}\,\widetilde{A}$.

Now $f_F = v_1 = \mathrm{id}_{\widetilde{A}}$, and hence
$$v_F(0) = 0_{\widetilde{A}} = (\varnothing, \varnothing).$$
On the other hand,
$$\begin{aligned}(\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0) &= \bigwedge Q \\ &= \bigwedge \mathrm{pt}\,\widetilde{A} \\ &= (\varnothing, I).\end{aligned}$$
Thus,
$$v_F(0) \neq (\mathcal{U}_* \circ [Q'] \circ \mathcal{U})(0),$$
so $\widetilde{A}$ is not stacked.

We next consider a stronger point-free version of the Hausdorff separation axiom. A frame A is said to be strongly Hausdorff, or (sH), if its codiagonal
$$\nabla_A : A \oplus A \longrightarrow A$$
is a closed quotient.

Unlike (H), this condition imposes a strong restriction on compact quotients, as the following result shows.

Proposition 3.17. *Every strongly Hausdorff frame is KC.*

Proof. This follows directly from [5, Theorem 9.2.1], which states that every compact quotient of a strongly Hausdorff frame is closed. □

Proposition 3.18. *Every KC frame is T_1.*

Proof. Let $p \in \operatorname{pt} A$. Since the completely prime filter F_p is open, A_{w_p} is a compact quotient of A. By the KC property, there exists $a \in A$ such that $w_p = u_a$.

Evaluating at 0, we obtain

$$a = u_a(0) = w_p(0) = p,$$

and hence $w_p = u_p$.

If $p < b \leq 1$, then $b \not\leq p$, so

$$1 = w_p(b) = u_p(b) = p \vee b = b.$$

Thus p is maximal in A. Since p was arbitrary, A is T_1. □

We briefly recall two separation properties that will be useful in what follows. A frame A is said to be *fit* if, whenever $a \not\leq b$, there exist $x, y \in A$ such that

$$a \vee x = 1, \qquad y \not\leq b, \qquad x \wedge y \leq b.$$

On the other hand, A is *subfit* if, whenever $a \not\leq b$, there exists $c \in A$ such that

$$a \vee c = 1, \qquad b \vee c \neq 1.$$

Every fit frame is subfit. We denote these properties by $(\mathbf{fit})$ and $(\mathbf{sfit})$, respectively.

Corollary 3.19. *Every subfit KC frame is spatial.*

Proof. By Proposition 3.18, every KC frame is T_1. The conclusion follows from [5, Proposition 2.5.3], since every subfit T_1 frame is spatial. □

Theorem 3.20. *There exists a KC frame which is strongly stacked and not subfit. Consequently,*

$$\mathrm{KC} \not\subseteq (\mathrm{sfit}).$$

Proof. Consider the frame $A = B \times 2$ introduced in Example 3.13. As shown there, A is T_1, strongly stacked, and non-spatial. Moreover,

$$A^\wedge = \{A, F_p\},$$

and

$$v_F = w_F \qquad \text{for every } F \in A^\wedge.$$

Thus, every nucleus in an admissibility interval $[v_F, w_F]$ is closed, so A is KC.

Finally, A cannot be subfit, since every subfit KC frame is spatial by Corollary 3.19. Therefore, A is KC, strongly stacked, and not subfit. □

3.2. **Inheritance and preservation properties.** The notions of stacked and strongly stacked frames are formulated in terms of the admissibility intervals associated with Hofmann–Mislove pairs. A related comparison was established by Simmons [9, Lemma 8.9 and Corollary 8.10]: for every H–M pair (F, Q),

$$\mathcal{U} \circ v_F \leq v_G \circ \mathcal{U}$$

where $G = \nabla(Q)$ and $\mathcal{U}$ is the spatial reflection.

We extend this inequality to arbitrary frame morphisms. Let $h : A \to B$ be a frame morphism, $G \in B^\wedge$, and put

$$F = h_* G \in A^\wedge.$$

The following lemmas establish the comparison between the corresponding prenuclei and their fitted nuclei.

Lemma 3.21. *Let $h : A \to B$ be a frame morphism, $G \in B^\wedge$, and $F = h_* G$. Then*

$$h \circ f_F \leq f_G \circ h.$$

Proof. Since

$$F = \{a \in A \mid h(a) \in G\},$$

for every $a \in F$ and $x \in A$ we have

$$h(v_a(x)) = h(a \succ x) \leq h(a) \succ h(x) = v_{h(a)}(h(x)) \leq f_G(h(x)).$$

Taking joins over $a \in F$ and using join preservation yields

$$h(f_F(x)) \leq f_G(h(x)).$$

□

Lemma 3.22. *Under the hypotheses of the preceding lemma, for every ordinal α,*

$$h \circ f_F^\alpha \leq f_G^\alpha \circ h.$$

In particular,

$$h \circ v_F \leq v_G \circ h.$$

Proof. We argue by transfinite induction on α. The case $\alpha = 0$ is immediate. If the inequality holds for α, then

$$\begin{aligned} h \circ f_F^{\alpha+1} &= h \circ f_F \circ f_F^\alpha \\ &\leq f_G \circ h \circ f_F^\alpha \\ &\leq f_G \circ f_G^\alpha \circ h \\ &= f_G^{\alpha+1} \circ h. \end{aligned}$$

For a limit ordinal λ, preservation of arbitrary joins gives

$$\begin{aligned} h(f_F^\lambda(x)) &= \bigvee_{\alpha<\lambda} h(f_F^\alpha(x)) \\ &\leq \bigvee_{\alpha<\lambda} f_G^\alpha(h(x)) \\ &= f_G^\lambda(h(x)). \end{aligned}$$

Finally, taking an ordinal at which both iterations have stabilized yields

$$h \circ v_F \leq v_G \circ h.$$

□

Proposition 3.23. *If A is a neat frame, then A_j is neat for every $j \in NA$.*

Proof. Suppose that A is α-neat and let $j \in NA$. Let $G \in A_j^\wedge$ and put $F = j_* G \in A^\wedge$. Set

$$\widehat{d} = f_F^\alpha(0), \qquad d = f_G^\alpha(0).$$

By Lemma 3.22, applied to $j^* : A \to A_j$, we have

$$j(\widehat{d}) = j(f_F^\alpha(0)) \leq f_G^\alpha(j(0)) = d.$$

Since j is inflationary,

$$\widehat{d} \leq j(\widehat{d}) \leq d.$$

Now let $y \in G$. Since $y \in A_j$, we have $j(y) = y$, and therefore $y \in F$. As A is α-neat,

$$y \vee \widehat{d} = 1.$$

Consequently,

$$1 = y \vee \widehat{d} \leq y \vee d \leq 1.$$

Thus A_j is α-neat, and hence neat. □

We now turn to coproducts. The comparison established in Lemma 3.22 allows us to prove that neatness is preserved under arbitrary coproducts, without first treating the binary case.

Theorem 3.24. *Neatness is preserved under arbitrary coproducts.*

Proof. Let $\{A_i\}_{i \in I}$ be a family of neat frames, and let α_i be a neatness degree of A_i. Put

$$A = \bigoplus_{i \in I} A_i, \qquad \alpha = \sup_{i \in I} \alpha_i,$$

and denote the canonical frame morphisms by $\iota_i : A_i \to A$.

Let $F \in A^\wedge$ and set

$$F_i = (\iota_i)_*[F] \in A_i^\wedge, \qquad d_i = f_{F_i}^\alpha(0), \qquad d = f_F^\alpha(0).$$

Since every A_i is α-neat,

$$x_i \vee d_i = 1 \qquad (x_i \in F_i).$$

By Lemma 3.22,

$$\iota_i(d_i) \leq f_F^\alpha(\iota_i(0)) \leq d.$$

Hence

$$\bigvee_{i \in I} \iota_i(d_i) \leq d.$$

Now let $x \in F$. Using the standard representation of the coproduct, write

$$x = \bigvee_{k \in K} \bigwedge_{i \in J_k} \iota_i(x_{k,i}),$$

where each $J_k \subseteq I$ is finite. Since F is completely prime, there is some $k \in K$ such that

$$\bigwedge_{i \in J_k} \iota_i(x_{k,i}) \in F.$$

Thus $x_{k,i} \in F_i$ for every $i \in J_k$. Consequently,

$$\begin{aligned} 1 &= \bigwedge_{i \in J_k} \big(\iota_i(x_{k,i}) \vee \iota_i(d_i)\big) \\ &\leq \bigwedge_{i \in J_k} \big(\iota_i(x_{k,i}) \vee d\big) \\ &= \left(\bigwedge_{i \in J_k} \iota_i(x_{k,i})\right) \vee d \\ &\leq x \vee d. \end{aligned}$$

Therefore $x \vee d = 1$ for every $x \in F$, so A is α-neat, and hence neat. □

The preceding results have a categorical consequence. Indeed, the subcategory of neat frames is closed under quotients, by Proposition 3.23, and under arbitrary coproducts, by Proposition 3.24. Hence, the standard criterion for coreflectivity of full subcategories of Frm applies.

Corollary 3.25. *The subcategory of neat frames is a coreflective subcategory of* Frm.

We now examine the inheritance of the stronger properties introduced above. We first show that strong stackedness is inherited by open quotients, and then establish the corresponding result for KC.

Recall that, for a frame morphism $f : A \to B$ and $j \in NA$, the induced nuclei satisfy

$$N(f)(j) = \bigvee_{a \in A} \big(u_{f(j(a))} \wedge v_{f(a)}\big), \qquad N(f_*)(k) = f_* \circ k \circ f.$$

Proposition 3.26. *Let A be a strongly stacked frame and $j \in NA$. Then A_j is strongly stacked.*

Proof. Let $F \in A_j^{\wedge}$, and put

$$G = j_* F, \qquad Q = \mathrm{pt}(A_j) \setminus F, \qquad R = \mathrm{pt}\, A \setminus G.$$

Let $\mathcal{U} \colon A \to \mathcal{O}(\mathrm{pt}\, A)$ and $\mathcal{U}_j \colon A_j \to \mathcal{O}(\mathrm{pt}\, A_j)$ be the corresponding spatial reflections. Set

$$l = N(\mathcal{U}_j)([Q']), \qquad m = N(j^*)(v_G).$$

By the general comparison between the admissibility bounds,

$$v_F \leq l.$$

We prove that

$$l \leq m \leq v_F.$$

Since $G = j_*F$, by Lemma 3.22,

$$j^* \circ v_G \leq v_F \circ j^*.$$

Consequently,

$$\begin{aligned} m &= \bigvee_{x \in A} \big(u_{j^*(v_G(x))} \wedge v_{j^*(x)}\big) \\ &\leq \bigvee_{x \in A} \big(u_{v_F(j^*(x))} \wedge v_{j^*(x)}\big) \leq v_F. \end{aligned}$$

Since A is strongly stacked,

$$v_G = N(\mathcal{U})([R']).$$

Hence

$$m = N(j^*)\big(N(\mathcal{U})([R'])\big).$$

Consider the auxiliary nucleus

$$n = N(\mathcal{U}_j \circ j^*)([Q']).$$

Using the explicit formula for the induced nucleus and the identification $Q = R \cap \mathrm{pt}(A_j)$, one obtains

$$j^*(n(x)) = l(x) \qquad (x \in A_j).$$

Moreover, the description of the right adjoint j_* gives

$$j^*(n(x)) \leq j^*\big(N(\mathcal{U})([R'])(x)\big).$$

Therefore,

$$u_{j^*(n(x))} \wedge v_x \leq u_{j^*(N(\mathcal{U})([R'])(x))} \wedge v_x \leq m.$$

Taking joins over $x \in A_j$ yields $l \leq m$. Thus,

$$v_F \leq l \leq m \leq v_F,$$

and consequently

$$v_F = N(\mathcal{U}_j)([Q']).$$

Therefore A_j is strongly stacked. □

In particular, since every strongly stacked frame is stacked, we also obtain that A_j is stacked.

Proposition 3.27. *Let A be a KC frame and $j \in NA$. Then A_j is KC.*

Proof. Let $k \in N(A_j)$ be such that $(A_j)_k$ is compact. By the correspondence of open filters, the nucleus

$$l = j_* \circ k \circ j^*$$

defines a compact quotient A_l. Since A is KC, $l = u_a$ for some $a \in A$.

For every $x \in A$,

$$a \vee x = l(x) = j_*(k(j^*(x))).$$

Applying j^* and using $j^* \circ j_* = \mathrm{id}_{A_j}$, we obtain

$$j^*(a) \vee j^*(x) = k(j^*(x)).$$

Since j^* is surjective, it follows that

$$k = u_{j^*(a)}.$$

Thus $(A_j)_k$ is closed, and hence A_j is KC. □

We conclude this section by comparing the compact quotients of a frame with those of its quotients. Let $j \in NA$, $F \in A^\wedge$, and $G = j_*F$. Consider the compact quotients

$$A_F = \{x \in A \mid v_F(x) = x\}, \qquad A_{j,G} = \{y \in A_j \mid v_G(y) = y\}.$$

By Lemma 3.22,

$$j \circ v_F \leq v_G \circ j.$$

Since v_G is idempotent, this implies

$$v_G \circ j \circ v_F = v_G \circ j.$$

Thus, the frame morphism $H = v_G \circ j : A \to A_{j,G}$ factors through v_F, inducing

$$h : A_F \longrightarrow A_{j,G}, \qquad h(x) = v_G(j(x)).$$

Proposition 3.28. *The map h is a frame morphism, and the following diagram commutes:*

$$\begin{array}{ccc} A & \xrightarrow{v_F} & A_F \\ {\scriptstyle j}\downarrow & & \downarrow{\scriptstyle h} \\ A_j & \xrightarrow[v_G]{} & A_{j,G}. \end{array} \tag{5}$$

Proof. The map h preserves finite meets and 1, as it is the restriction of $H = v_G \circ j$.

Let $X \subseteq A_F$. Since joins in A_F are computed by applying v_F, we have

$$\begin{aligned} h\left(\bigvee_{A_F} X\right) &= v_G\left(j\left(v_F\left(\bigvee_{A} X\right)\right)\right) \\ &= v_G\left(j\left(\bigvee_{A} X\right)\right) \\ &= \bigvee_{A_{j,G}} h[X]. \end{aligned}$$

Thus h preserves arbitrary joins. Finally,

$$h \circ v_F = v_G \circ j$$

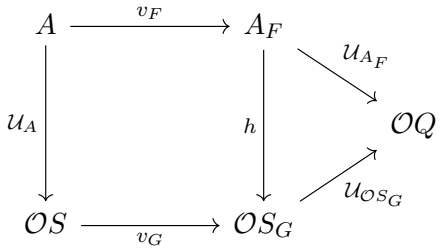
by construction, so the diagram commutes. □

We now specialize the preceding construction to the spatial reflection $\mathcal{U}_A : A \to \mathcal{O}(\mathrm{pt}\, A)$. Let $F \in A^\wedge$, and let $Q \subseteq \mathrm{pt}\, A$ be its associated compact saturated subset. By the Hofmann–Mislove correspondence, put $G = \nabla(Q)$.

The point spaces of A_F and $(\mathcal{O}(\mathrm{pt}\, A))_G$ are both identified with Q. Consequently, the preceding construction yields a commutative diagram relating these compact quotients to $\mathcal{O}Q$.

By [8, Lemma 5.7.1], the point space of the compact quotient A_F is precisely Q. Consequently, the preceding constructions fit into the following commutative diagram:

(6)



Since this diagram is determined by $F \in A^\wedge$ and its associated compact saturated subset $Q \in \mathcal{Q}S$, we shall refer to it as the Q-square.

Theorem 3.29. *Under the hypotheses of the Q-square, if A satisfies (H), then*

$$\mathcal{O}Q \cong Q'.$$

Equivalently, the frame of opens of the point space of A_F is isomorphic to a compact closed quotient of a Hausdorff space.

Proof. Suppose that A satisfies (H). Since (H) is conservative, $S = \operatorname{pt} A$ is Hausdorff. Thus Q is closed in S, and hence $Q' \in \mathcal{O}S$.

Since every Hausdorff space is strongly stacked,

$$v_G = [Q'] = u_{Q'}.$$

Therefore,

$$(\mathcal{O}S)_{v_G} = (\mathcal{O}S)_{u_{Q'}} \cong Q'.$$

Moreover, $u_{Q'}$ is a complemented nucleus, so $(\mathcal{O}S)_{v_G}$ is spatial. Since

$$\operatorname{pt}((\mathcal{O}S)_{v_G}) = Q,$$

we conclude that

$$\mathcal{O}Q \cong (\mathcal{O}S)_{v_G} \cong Q'.$$

□

4. Interaction among compact-quotient separation properties

We begin with a result of Sexton [6, Theorem 6.2(a)], which states that, for every topological space S,

$$\mathcal{O}S \text{ neat} \implies S \text{ strongly stacked} \implies S \text{ stacked}.$$

This provides a natural starting point for comparing neatness and strong stackedness in the point-free setting.

Proposition 4.1. *If A is a neat frame, then $\operatorname{pt} A$ is strongly stacked.*

Proof. Let $S = \operatorname{pt} A$. Since neatness is hereditary, $\mathcal{O}S$ is neat. By [6, Theorem 6.2(a)], S is strongly stacked. □

The preceding result concerns the point space of A, rather than A itself. We next identify a condition under which neatness implies strong stackedness of the frame.

Proposition 4.2. *Let A be a neat frame. If*

$$\mathcal{U} \circ v_F = v_G \circ \mathcal{U}$$

for every H–M triple (F, Q, G), then A is strongly stacked.

Proof. Let $S = \operatorname{pt} A$ and let (F, Q, G) be an H–M triple. Since A is neat, $\mathcal{O}S$ is neat, and hence S is stacked and packed. Thus Q is closed in S, so $Q' \in \mathcal{O}S$.

By stackedness and Lemma 3.4,

$$v_G(0) = Q'.$$

Since $\mathcal{O}S$ is neat,

$$v_G = u_{Q'} = [Q'].$$

By hypothesis,

$$\mathcal{U} \circ v_F = [Q'] \circ \mathcal{U}.$$

Applying the right adjoint $\mathcal{U}_*$, we obtain

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Therefore A is strongly stacked. □

We next compare KC with the preceding properties.

Proposition 4.3. *Every KC frame is neat.*

Proof. Let $F \in A^\wedge$. Since v_F belongs to the admissibility interval $[v_F, w_F]$ and A is KC, there exists $d \in A$ such that

$$v_F = u_d, \qquad d = v_F(0).$$

Thus A is neat. □

Proposition 4.4. *Let A be a KC frame. Then A is stacked if and only if it is strongly stacked.*

Proof. Since every KC frame is T_1 and neat, for each $F \in A^\wedge$ there exist $d, a \in A$ such that

$$v_F = u_d, \qquad w_F = u_a,$$

where $d = v_F(0)$ and $a = w_F(0)$.

By Proposition 3.9,

$$\begin{aligned} A \text{ stacked} &\iff v_F(0) = w_F(0) \quad \text{for every } F \\ &\iff d = a \quad \text{for every } F \\ &\iff v_F = w_F \quad \text{for every } F \\ &\iff A \text{ strongly stacked.} \end{aligned}$$

□

Proposition 4.5. *Let A be a spatial frame. Then A is KC if and only if it is neat.*

Proof. The forward implication follows from Proposition 4.3. Conversely, suppose that A is spatial and neat. Write $A \cong \mathcal{O}S$ for some topological space S. By the spatial characterization of neatness, S is packed and stacked, so S is strongly stacked. Hence $\mathcal{O}S$ is KC, and therefore so is A. □

Proposition 4.6. *Every neat strongly stacked frame is KC.*

Proof. Let A be neat and strongly stacked. Since neatness implies T_1, by Proposition 3.9,

$$v_F = w_F \qquad (F \in A^\wedge).$$

Moreover, by neatness,

$$v_F = u_{v_F(0)}.$$

Thus every admissibility interval $[v_F, w_F]$ consists of a single closed nucleus. Hence A is KC. □

We have shown that every KC frame is neat and that every neat strongly stacked frame is KC. This naturally raises the question of whether strong stackedness is necessary, or equivalently, whether neatness alone implies the KC property.

An affirmative answer would make neatness and KC equivalent for arbitrary frames, as already happens in the spatial setting. The following example shows that this is not the case.

Theorem 4.7. *There exists a non-spatial neat frame which is neither KC nor strongly stacked. Consequently,*

$$\mathrm{KC} \subsetneq \mathrm{Neat}, \qquad \mathrm{Neat} \not\subseteq \mathrm{S.S.}$$

Proof. Let B be a nontrivial atomless complete Boolean algebra, and consider the frame

$$C = \{(a, b) \in B \times B : a \leq b\},$$

with coordinatewise operations. Let

$$A = C \cup \{\top\},$$

where $\top$ is a new greatest element, so that $1_C = (1_B, 1_B) < \top$. We shall prove that A is neat but not KC.

%textitStep 1. The Scott-open filters of A.

We first show that the only Scott-open filter of B is B itself.

Indeed, let F be a Scott-open filter of B. Suppose that $0_B \notin F$. For every $a \in F$, we have $a > 0_B$. Since B is atomless, we can choose b_a such that

$$0_B < b_a < a.$$

In particular, a is not compact. More generally, a complete atomless Boolean algebra has no nonzero compact elements.

Now put

$$d = \bigwedge F.$$

The family

$$D = \{d \vee \neg a : a \in F\}$$

is directed, since F is closed under finite meets. Moreover, by complete distributivity of finite meets over arbitrary joins,

$$\bigvee D = d \vee \bigvee_{a \in F} \neg a = d \vee \neg d = 1_B.$$

Since $1_B \in F$ and F is Scott-open, there exists $a \in F$ such that

$$d \vee \neg a \in F.$$

Consequently,

$$d = a \wedge (d \vee \neg a) \in F.$$

Thus $F = {\uparrow} d$. Scott-openness now implies that d is compact, and hence $d = 0_B$. Therefore $F = B$, proving that

$$B^\wedge = \{B\}.$$

We next determine the Scott-open filters of A. Let $G \in A^\wedge$.

If $G \cap C \neq \varnothing$, then $1_C \in G$. Consider the frame homomorphism

$$\delta : B \longrightarrow C, \qquad \delta(b) = (b, b).$$

Since the inclusion $C \hookrightarrow A$ preserves directed joins, the set

$$H = \{b \in B : (b, b) \in G\}$$

is a Scott-open filter of B. Hence $H = B$. In particular, $(0_B, 0_B) \in G$, so $G = A$.

On the other hand, if $G \cap C = \varnothing$, then necessarily $G = \{\top\}$.

Finally, $\{\top\}$ is Scott-open, because $\top$ is compact in A: any family of elements whose join is $\top$ must contain $\top$.

We conclude that

$$A^\wedge = \{A, \{\top\}\}.$$

Recall that, for every Scott-open filter F, the associated fitted nucleus is

$$v_F = \left(\bigvee_{a \in F} v_a \right)^\infty.$$

Since $0_A \in A$, we have

$$v_A = v_{0_A} = \mathrm{tp} = u_\top.$$

Similarly, since $v_\top = \mathrm{id}_A$,

$$v_{\{\top\}} = v_\top = \mathrm{id}_A = u_{0_A}.$$

Thus v_F is a closed nucleus for every $F \in A^\wedge$. Consequently, A is neat.

Let $e = (0_B, 1_B) \in C$. The open nucleus v_e of C is given by

$$v_e(a, b) = e \succ_C (a, b) = (b, b).$$

Indeed, for $(c, d) \in C$,

$$(c, d) \wedge e = (0_B, d),$$

and therefore

$$(c, d) \wedge e \leq (a, b) \quad \iff \quad d \leq b.$$

Define $j : A \to A$ by

$$j(x) = \begin{cases} (b,b), & x = (a,b) \in C, \\ \top, & x = \top. \end{cases}$$

The map j is inflationary and idempotent. Moreover, for $(a,b),(c,d) \in C$,

$$\begin{aligned} j((a,b) \wedge (c,d)) &= j(a \wedge c, b \wedge d) \\ &= (b \wedge d, b \wedge d) \\ &= j(a,b) \wedge j(c,d). \end{aligned}$$

The cases involving $\top$ are immediate. Hence j is a nucleus on A.

Its fixed-point frame is

$$A_j = \{(b,b) : b \in B\} \cup \{\top\}.$$

We claim that A_j is compact. Indeed, suppose that

$$\bigvee_{A_j} x_i = \top$$

for some family $\{x_i\}_{i \in I} \subseteq A_j$. If none of the x_i were equal to $\top$, then $x_i = (b_i, b_i)$ for every i, and hence

$$\begin{aligned} \bigvee_{A_j} x_i &= j\left(\bigvee_{A} x_i\right) \\ &= j\left(\bigvee_i b_i, \bigvee_i b_i\right) \\ &= \left(\bigvee_i b_i, \bigvee_i b_i\right) \\ &\leq 1_C < \top, \end{aligned}$$

a contradiction. Thus some $x_i = \top$, which proves that A_j is compact.

However,

$$j(0_A) = 0_A,$$

whereas

$$j(0_B, 1_B) = (1_B, 1_B) \neq (0_B, 1_B).$$

Therefore $j \neq \mathrm{id}_A$.

If j were closed, then

$$j = u_{j(0_A)} = u_{0_A} = \mathrm{id}_A,$$

a contradiction. Hence j is not closed.

We have thus constructed a compact quotient A_j of A which is not closed. Therefore A is not KC.

Consequently, there exists a neat frame which is not KC. Since every KC frame is neat, we obtain

$$\mathrm{KC} \subsetneq \mathrm{Neat}.$$

Finally, since every neat strongly stacked frame is KC, the frame A is not strongly stacked either. □

The preceding theorem shows that neatness alone does not imply either KC or strong stackedness. We now examine additional separation conditions under which these implications can be recovered.

Proposition 4.8. *Let A be a KC frame. If A is subfit, then A is strongly stacked.*

Proof. Let (F, Q) be an HM-pair and put

$$d = v_F(0) \qquad \text{and} \qquad a = w_F(0).$$

Since A is KC, both v_F and w_F are closed nuclei. Hence

$$v_F = u_d \qquad \text{and} \qquad w_F = u_a.$$

By subfitness and [10, Theorem 3.5], we have

$$d = a.$$

Therefore,

$$v_F = u_d = u_a = w_F.$$

Since A is T_1, we also have

$$w_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Consequently,

$$v_F = \mathcal{U}_* \circ [Q'] \circ \mathcal{U}.$$

Since this holds for every HM pair (F, Q), A is strongly stacked. □

Proposition 4.9. *Let A be a frame. If A is neat and fit, then A is KC.*

Proof. Let $F \in A^\wedge$ and $j \in NA$ be such that

$$\nabla(j) = F.$$

Since A is fit, by [10, Theorem 3.4], every nucleus on A is fitted. Hence j is the least nucleus in its block. On the other hand, v_F is also the least nucleus admitting F. Therefore,

$$j = v_F.$$

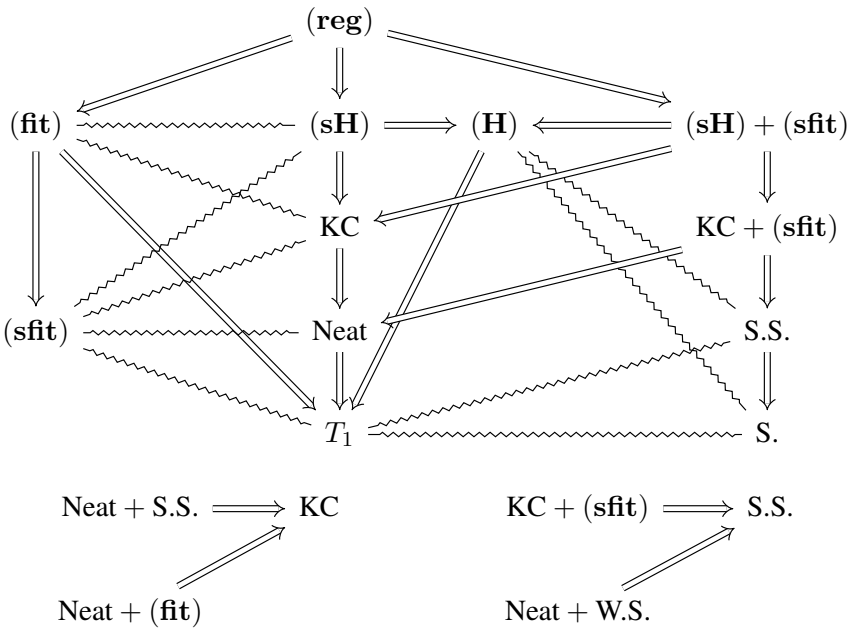
Since A is neat, there exists $d \in A$ such that

$$v_F = u_d.$$

Consequently,

$$j = u_d.$$

Thus every nucleus whose associated filter is open is closed, and hence A is KC. □



5. Relationship with the patch assembly

Following the idea presented in [10] (or, alternatively, in [9]), let us consider the linearly ordered frame

$$A = [0, 1].$$

In this case the following observations hold.

(1) For all $a, b \in A$ the implication $(a \succ b)$ is given by

$$(a \succ b) = \begin{cases} 1, & \text{si } a \leq b, \\ b, & \text{si } a > b. \end{cases}$$

(2) For $a, x \in A$, the distinguished nuclei u_a, v_a, w_a are given by

$$u_a(x) = \begin{cases} x & \text{si } a \leq x, \\ a & \text{si } x < a, \end{cases}, \quad v_a(x) = \begin{cases} 1 & \text{si } x \geq a, \\ a & \text{si } x < a, \end{cases} \quad w_a(x) = \begin{cases} 1 & \text{si } x > a, \\ a & \text{si } x \leq a. \end{cases}$$

(3) If $F \subseteq A$ is a filter, then F is always admissible and has the form

$$F = \nabla(w_a) = (a, 1] \quad \text{o} \quad F = \nabla(v_a) = [a, 1]$$

for some $a \in A$. In particular, $F \in A^\wedge$ if and only if $F = (a, 1]$.

(4) If $F \in A^\wedge$, the adjusted nucleus associated v_F is calculated as

$$v_F(x) = l_m(x) = \begin{cases} 1 & \text{if } x > m, \\ x & \text{if } x \leq m. \end{cases} \tag{7}$$

where $l_m = v_m \wedge w_m$ and $m = \bigwedge F$.

(5) For $S = \operatorname{pt} A$ it holds that $S = [0, 1)$ and the open sets in $\mathcal{O}S$ are of the form
$$\mathcal{U}_A(x) = \{p \in S \mid p < x\} = [0, x).$$
(6) The frame A is not T_1, since no $p \in S$ is maximal. Consequently, A does not satisfy any stronger separation property in Frm.
(7) For $F \in A^\wedge$ the compact saturated set associated $Q \in \mathcal{Q}S$ is of the form $Q = [0, m]$.
(8) It holds that $[Q'] = v_F$; therefore, S is strongly stacked.
(9) Since $[0, x) \in \mathcal{O}S$, then $[x, 1] \in \mathcal{C}S$ and $Q = [0, m]$. Therefore, the compact subsets are not closed; that is, S is not packed.
(10) The frame A is not efficient.

The above resume some characteristics about the frame $[0, 1]$. Now, we will perform the analysis of the patch constructions of A and its point space S.

Remember that for $A \in \mathrm{Frm}$ and $S \in \mathrm{Top}$ we have

$$\text{Pbase} = \{u_a \wedge v_F \mid a \in A, F \in A^\wedge\} \quad \text{y} \quad \text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}$$

where Pbase generates the patch frame of A and pbase generates the topology of the patch space of S. By (7) we have that

$$\text{Pbase} = \{u_a \wedge l_m \mid a \in [0, 1], m \in [0, 1)\}.$$

For $x \in A$ we have

$$(u_a \wedge l_m)(x) = u_a(x) \wedge l_m(x).$$

Let us analyze the value of this map according to the relative position of x and m.

Case 1: If $x < m$, then
$$(u_a \wedge l_m)(x) = u_a(x) \wedge x = x.$$

Case 2: If $x = m$, then
$$(u_a \wedge l_m)(x) = u_a(m) \wedge m = x.$$

Case 3: If $x > m$, then
$$(u_a \wedge l_m)(x) = u_a(x) \wedge 1 = u_a(x).$$

In the first two cases, the value of a does not affect the evaluation. Thus, the only non-trivial case occurs when $m < x$, where the value of $(u_a \wedge l_m)(x)$ depends on the relative position of a and x. We therefore distinguish the following two subcases.

Case 3.1: If $m < x < a$, then
$$(u_a \wedge l_m)(x) = a.$$

Case 3.2: If $m < a \leq x$, then
$$(u_a \wedge l_m)(x) = x.$$

From the preceding analysis, we conclude that the non-trivial generators of Pbase correspond precisely to the pairs (m, a) with $m < a$. Consequently, the frame PA is determined by the open intervals of the form (m, a), where $m \in \mathrm{pt}\, A$ and $a \in A$.

We now turn to the question of whether PA satisfies any separation property. To this end, we recall the relationship between the point-free patch construction and its spatial counterpart.

Let A be an arbitrary frame and let $S = \mathrm{pt}\, A$ be its space of points. The two patch constructions are related by the following commutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & PA & \hookrightarrow & NA \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle P(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\ \mathcal{O}S & \longrightarrow & P\mathcal{O}S & \hookrightarrow & N\mathcal{O}S \\ & \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

where $\mathcal{O}^f S$ is the Skula topology (or front topology) of S. In [7] is named *the patch assembly diagram* and more details about it can also be found in [**?**].

In our case, $\mathcal{U}_A$ is an isomorphism. Moreover, under $\mathcal{U}_A$, the patch construction turns out to be functorial and, therefore, $PA \cong P\mathcal{O}S$. Similarly, $NA \cong N\mathcal{O}S$. To describe explicitly $P\mathcal{O}S$ and $N\mathcal{O}S$, we use the following result, which can be consulted in [1].

Proposition 5.1. *Let S be a partially ordered set.*

(1) *If S has no infinite antichains, then $N\mathcal{O}S$ is spatial.*
(2) *If S is totally ordered, then $N\mathcal{O}S$ is spatial.*

Thus, by Proposition 5.1 we have that $N\mathcal{O}S$ is spatial and therefore $N\mathcal{O}S \cong \mathcal{O}^f S$. With all this information, the patch assembly diagram, in our particular case is as follows.

$$\begin{array}{ccccc} A & \longrightarrow & PA & \longrightarrow & NA \\ {\scriptstyle \cong}\downarrow & & \downarrow{\scriptstyle \cong} & & \downarrow{\scriptstyle \cong} \\ \mathcal{O}S & \longrightarrow & \mathcal{O}^p S & \longrightarrow & \mathcal{O}^f S \end{array}$$

Therefore, if we want to study the frame PA, we can study the patch topology $\mathcal{O}^p S$. Remember that

$$\text{pbase} = \{U \cap Q' \mid U \in \mathcal{O}S, Q \in \mathcal{Q}S\}.$$

In our case,

$$\text{pbase} = \{[0, a) \cap [m, 1) \mid a \in [0, 1], m \in [0, 1)\} = \{(m, a)\}.$$

Since the topology generated by the intervals (m, a) coincides with the usual topology on $[0, 1)$, the space ${}^p S$ is Hausdorff. Consequently, $\mathcal{O}^p S \cong PA$ satisfies $(\mathbf{H})$, even when A is not T_1 nor neat.

Lemma 5.2. *Let A be a frame, if the morphism $\mathrm{p}U \colon \mathrm{p}A \to \mathrm{p}\mathcal{O}S$ is an isomorphism, then A is strongly stacked.*

Proof. Consider any open filter F on A and the compact saturated determined by it, say Q, thus we have the nuclei v_F and $U_*[Q']U = Q[U]$, we neee to observe the action of $\mathrm{p}U$ on these two nuclei, in order to do that remember that we have a conmutative diagram:

$$\begin{array}{ccccc} A & \longrightarrow & \mathrm{p}A & \xhookrightarrow{\iota} & NA \\ {\scriptstyle \mathcal{U}_A}\downarrow & & \downarrow{\scriptstyle \mathrm{p}(\mathcal{U}_A)} & & \downarrow{\scriptstyle N(\mathcal{U}_A)} \\ \mathcal{O}S & \longrightarrow & \mathrm{p}\mathcal{O}S & \xhookrightarrow{\imath} & N\mathcal{O}S \\ & \searrow & \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \sigma} \\ & & \mathcal{O}^p S & \hookrightarrow & \mathcal{O}^f S \end{array}$$

since $N(\mathcal{U})\iota = \imath \mathrm{p}\mathcal{U}$, that is, $\mathrm{p}\mathcal{U}$ is the restriction of $N(\mathcal{U})$ to $\mathrm{p}A$, then

$$\mathrm{p}\mathcal{U}(\mathrm{v}_F) = N(\mathcal{U})(\mathrm{v}_F) = \bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

since the morphism $\mathcal{U}$ transforms open filters we have

$$\bigvee\{u_{\mathcal{U}(v_F)} \wedge v_{\mathcal{U}(x)} \mid x \in A\} = \bigvee\{u_{v_{\mathcal{U}F}} \wedge v_{\mathcal{U}(x)} \mid x \in A\}$$

But this representation is the corresponding to the nucleus $\mathrm{v}_{\mathcal{U}F}$.

Completar esta prueba

□

Considerar incluir información sobre los parches

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